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1. Comparison of L1, L2, and L3 Cache Levels

Parameter	L1 Cache	L2 Cache	L3 Cache
Typical Size	\$32\text{ KB} - 64\text{ KB}\$ per core	\$512\text{ KB} - 1\text{ MB}\$ per core	\$8\text{ MB} - 64\text{ MB}\$ (Shared)
Access Latency	\$0.5 - 1.5\text{ ns}\$ (\$4 - 5\text{ cycles}\$)	\$3 - 7\text{ ns}\$ (\$12 - 14\text{ cycles}\$)	\$10 - 25\text{ ns}\$ (\$40 - 75\text{ cycles}\$)
Bandwidth	Ultra-High (\$>2 - 5\text{ TB/s}\$)	High (\$1 - 2\text{ TB/s}\$)	Moderate-High (\$300 - 800\text{ GB/s}\$)
Throughput	2–3 memory ops / cycle	1 memory op / cycle	Multi-channel interleaved
Placement & Sharing	Inside core pipeline; private	Near core; private/semi-shared	Outside cores; shared across all cores
Impact on Performance	Eliminates immediate ALU stalls during instruction loops.	Reduces penalties from L1 data/instruction evictions.	Prevents high-penalty off-chip DRAM bus round-trips.

2. Differentiation of SRAM vs. DRAM

Parameter	SRAM (Static RAM)	DRAM (Dynamic RAM)
Cell Structure	6 Transistors (6T configuration)	1 Transistor + 1 Capacitor (1T-1C)
Access Latency	Ultra-low ($0.5 - 5 \text{ ns}$)	Moderate-High ($50 - 80 \text{ ns}$)
Bandwidth	Terabytes per second (on-chip buses)	$30 - 100 \text{ GB/s}$ per channel (off-chip bus)
Throughput	Immediate cycle-by-cycle pipelining	Limited by row-buffer activations/precharge cycles
Refresh Need	None (holds data while powered)	Periodic refresh required (causes latency jitter)
Density & Cost	Low density, physically large, high cost	Very high density, compact footprint, low cost
Hierarchical Role	L1, L2, and L3 CPU/GPU Cache memories	Main system memory (Primary working memory)

3. Virtual Memory with Paging vs. Cache Memory

Parameter	Cache Memory Mechanism	Virtual Memory with Paging
Management	100% Hardware-controlled (Cache Controller)	Software + Hardware (OS + MMU + TLB)
Granularity	Cache line (64 bytes)	Page (4 KB , 2 MB , or 1 GB)
Access Latency	$1 - 25\text{ ns}$	100 ns (TLB hit) to $5 - 10\text{ ms}$ (Page Fault from disk)
System Throughput	Multi-terabyte sustained throughput	Lower throughput; page faults stall the active thread
Address Mapping	Direct-Mapped / Set-Associative via tags	Multi-level Page Tables + TLB translation
Miss Penalty	Fetching line from DRAM ($50 - 100\text{ ns}$)	Disk I/O interrupt and context switch ($\mu\text{s} - \text{ms}$)

4. NAND Flash vs. DRAM

Parameter	NAND Flash	DRAM
Volatility	Non-Volatile (retains data without power)	Volatile (loses data on power loss)
Access Latency	High: $25 - 100\ \mu\text{s}$ (Reads), $200 - 500\ \mu\text{s}$ (Writes)	Low: $50 - 80\ \text{ns}$ (Reads and Writes)
Bandwidth	$2 - 7\ \text{GB/s}$ (over PCIe/NVMe Gen 4/5)	$50 - 100\ \text{GB/s}$ per DDR5 channel
Access Mode	Block/Page addressed (no byte access)	Byte-addressable
Endurance	Finite ($10^3 - 10^5$ write cycles per cell)	Virtually unlimited ($>10^{15}$ write cycles)
Hierarchical Role	Secondary Storage (SSDs, mass file storage)	Primary Memory (active execution space)

5. MRAM vs. RRAM (Emerging NVM Technologies)

Parameter	MRAM (Magnetoresistive RAM)	RRAM (Resistive RAM)
Storage Mechanism	Magnetic Tunnel Junction (MTJ resistance)	Conductive filament formation in dielectric
Access Time / Latency	Read: $1 - 10\text{ ns}$, Write: $5 - 20\text{ ns}$	Read: $10 - 50\text{ ns}$, Write: $10 - 100\text{ ns}$
Throughput	Very High (near-SRAM execution speeds)	High for sequential reads; moderate writes
Energy Efficiency	Extremely low active and zero standby power	Low write energy; ideal for Analog Matrix ops
Write Endurance	Extremely high ($10^{12} - 10^{15}$ cycles)	Moderate to High ($10^6 - 10^{10}$ cycles)
Primary Suitability	High-speed cache replacement, instant-on embedded	Dense AI model storage, Compute-In-Memory (CiM)